Colter Snyder

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SUMMARY

Software engineer with two years of hands-on development on mission-critical defense simulation software in ITAR- and CUI-controlled environments. Works across the full software lifecycle in Python, Rust, and C++ on Linux, using GitLab CI/CD, test-driven development, and agile practices, and uses AI-assisted tools (Claude Code, Cursor) within export-control limits. M.S. in Computer Science (3.97 GPA); former adjunct instructor to more than 200 students; hobbyist in FPGA design and 6502/Z80 assembly.

EXPERIENCE

Innovative Scientific Solutions, Inc.
Software Engineer II — *Dec 2024 – Present*

- Develop ITAR- and CUI-controlled platforms for probabilistic nuclear weapons effects simulation, covering blast, neutron radiation, thermal fluence, ground shock, cratering, and X-ray effects, for defense and allied government agencies.
- Work across the full lifecycle on a 4–5 person agile team: refine requirements, write object-oriented designs, UML diagrams (StarUML), and design documentation, implement features, and verify them with automated tests.
- Port legacy physics models from Fortran and Python to Rust, preserving their numerical behavior.
- Build Python (Flask) and Rust back-end services with WebSocket communication and MongoDB, deployed with Docker in Linux environments.
- Spearheaded an internal Python performance-benchmarking tool, a CLI with a GUI front end, that benchmarks our calculation services by individual and combined effect and charts the results.
- Maintain code quality with Git, GitLab CI/CD, and SonarQube; build interfaces in Angular, TypeScript, and Tauri.
- Apply AI-assisted tools (Claude Code, Cursor) only to non-CUI, non-ITAR idea exploration and research, keeping all controlled code and data out of commercial AI services.

Tyler Technologies — Lakewood, CO
Software Development Intern — *May 2023 – Aug 2023*

- Overhauled a 2,000-test automated suite, restoring about 1,500 tests (75%) to passing and giving the team a reliable foundation for test-driven development.
- Migrated a major legacy VB6 desktop application to object-oriented C# on an agile cloud modernization team.

Colorado School of Mines — Golden, CO
Postgraduate Research Assistant, Department of Computer Science — *May 2024 – Dec 2024*

- Developed object detection models for common semiconductor cleanroom equipment and integrated them, with Python back-end services, into a Unity AR training app, delivering a working demo in collaboration with the University of Arizona (advisor: Dr. Mehmet Belviranli).

Undergraduate Research Fellow — *Sep 2020 – Apr 2021*

- Designed and built the front end and back end of a collaborative VR environment for engineering education; co-authored *Collaborative Environment for Education in VR* for a Mines research conference (2021).

TEACHING & MENTORING

Colorado School of Mines — Golden, CO
Adjunct Professor, Department of Computer Science — *Jan 2023 – Dec 2023*

- Co-taught CSCI 220: Data Structures and Algorithms (C++) to 120 undergraduates, coaching students on object-oriented design, debugging, and algorithmic problem solving (Fall 2023).
- Taught a section of CSCI 102: Intro to Computer Science Lab (Python) to 90 undergraduates (Spring 2023).

SKILLS

Languages: Python, C++, Rust, C#, Fortran, TypeScript, JavaScript, SQL

Practices: Object-Oriented Design, UML, Requirements & Design Documentation, Agile/Scrum, Test-Driven Development, CI/CD

Tools: Linux, Git, GitLab CI/CD, SonarQube, Docker, StarUML, MongoDB, Angular, Tauri, Claude Code, Cursor

Personal Hardware Projects: FPGA design (Verilog), 6502 & Z80 assembly programming, CAD/3D printing

EDUCATION

Colorado School of Mines — Golden, CO

Master of Engineering in Nuclear Engineering (in progress) — *Aug 2026 – Expected May 2031*
Master of Science in Computer Science • GPA 3.97 — *Jan 2022 – May 2024*
Bachelor of Science in Computer Science • Magna Cum Laude • GPA 3.77 — *Aug 2019 – Dec 2022*

Relevant coursework: Computer Graphics & Advanced Computer Graphics (C++), Cyber-Physical Systems Security

Certification: CAE-CDE Certificate, NSA/DHS National Center of Academic Excellence in Cyber Defense Education

Honors: CACI C-MAPP Scholar, National Merit Scholar, Dean's List, Eagle Scout, Rotary Youth Leadership Award